

A variational graph learning framework for interpretable phenotype-based gene pathway activity identification of spatial transcriptomics

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Abstract. Spatial transcriptomics (ST) provides powerful insights into gene expression within tissue structures, enabling the discovery of molecular mechanisms in complex tumor microenvironments (TMEs). Despite advances in computational methods for ST analysis, existing tools lack the ability to simultaneously link specific tissue regions with clinically relevant phenotypes *and* identify the gene pathways that mechanistically govern those associations at the spatial level. Recent methods that bridge ST and clinical data either rely on rare matched spatial-clinical cohorts, restrict their analysis to transcription factor activities or cell-type compositions, or perform global bulk-based decompositions that do not explicitly model spatial tissue organization. Here, we present **GRAPHIST** (**GR**aph learning method for **PH**enotype-based **Inter**pretable pathway activity identification of **S**patial **T**ranscriptomics), a graph learning framework that first integrates phenotype-associated bulk expression and ST data by measuring similarity between each spot and bulk sample, and then optimizes a regression model on the resulting correlation matrix to pinpoint tissue regions significantly associated with a phenotype of interest. Subsequently, a biologically interpretable variational graph autoencoder quantifies the activity of prior gene pathways in various tissue regions—specifically the phenotype-associated regions—enabling mechanistic interpretation beyond cell-type composition or transcription factor inference. When applied to pancreatic and breast cancer TME datasets, GRAPHIST successfully identified biologically relevant tissue regions tied to candidate biomarker gene knockdowns and survival outcomes, and pinpointed previously annotated pathways governing the corresponding phenotype associations in specific tissue spots, demonstrating its potential for uncovering pathway-level molecular insights within complex tissues.

Keywords: Spatial Transcriptomics · Gene Pathway · Graph Neural Networks

1 Introduction

The tumor microenvironment (TME) is a complex and dynamic ecosystem composed of a heterogeneous mixture of cell phenotypes, subtypes, and spatial structures. The highly heterogeneous tumor ecosystem contains not only malignant cells but also other cell types, such as endothelial cells, stromal fibroblasts, and a variety of immune cells whose cellular composition, molecular features, and spatial arrangement in the TME all contribute to how tumors evolve and respond to different therapeutic strategies. Thus, the TME plays a crucial role in cancer progression, metastasis, and resistance to therapy. For instance, the cellular composition, spatial organization of immune cells, and their interaction with non-immune cells within the TME can influence the effectiveness of immune responses, leading to varied treatment outcomes among patients [6, 7, 18].

Spatial transcriptomics (ST) has emerged as a powerful technique for studying cellular heterogeneity in the spatial context of TMEs by retaining spatial positions alongside measurements of transcriptomic gene expression at a variety of spatial locations (spots) in a tissue sample [21]. By maintaining spatial information, ST facilitates the exploration of how specific cell types and molecular pathways are organized within tissue architecture, providing a clearer picture of the TME’s functional landscape [8]. To identify spot clusters from ST data, the standard approach is to perform spatially aware unsupervised clustering methods, which incorporate spatial coordinates or distances between spots to define spot clusters using both gene expression similarity and physical proximity [28, 9, 27]. However, this unsupervised clustering of tissue spots is fundamentally limited in its ability to identify specific tissue domains linked to crucial clinical phenotypes, such as tumor progression, survival outcome, and response to treatment. Identification of phenotype-associated tissue regions is indispensable, since it will facilitate detection of prognostic markers specific to individual tissue domains [1]. Furthermore, ST technology is not yet practical in large cohorts due to the resource-intensive nature of collecting spatial data, leaving insufficient statistical power to robustly identify tissue domains associated with a phenotype of interest [11].

Meanwhile, extensive clinical phenotype information for large cohorts is readily available in publicly accessible databases such as The Cancer Genome Atlas (TCGA) [26], Gene Expression Omnibus (GEO) [3], and Genomics of Drug Sensitivity in Cancer (GDSC) [29]. The clinical phenotype information in these databases is primarily collected on bulk tissue samples. Consequently, linking bulk gene expression data and their corresponding phenotype information with ST data to identify phenotype-associated spots becomes imperative. Recently, single-cell RNA-sequencing-based methods such as Scissor [25], scAB [32], and PIPET [23] have advanced our understanding of single-cell subpopulations in relation to specific phenotypes measured on bulk samples. While these single-cell phenotype association tools have achieved significant success, they are not suitable for analyzing ST data because they disregard the spatial coordinates of cells or spots that are intrinsic to ST datasets.

Several recent computational frameworks have begun to address the challenge of linking ST data to clinical phenotypes. SpaLinker [2] decomposes bulk transcriptomic profiles into latent factors via non-negative matrix factorization, associates these factors with clinical outcomes, and maps their distributions back onto ST data. While SpaLinker can visualize inferred factors spatially, it performs global bulk-based factorization and does not explicitly model the spatial organization of the tissue—it identifies where factors are expressed, but does not leverage spot-level spatial structure in its phenotype association step. SpaPheno [5] bridges large bulk RNA-seq cohorts with ST data by projecting both into a shared cell-type feature space via deconvolution, then fitting a sparse Elastic Net regression with SHAP interpretability. Although SpaPheno provides multi-scale interpretability at the cell-type level and does not require matched spatial-clinical data, its feature representation is restricted to cell-type composition—it cannot attribute phenotype associations to specific gene pathways or molecular programs operating within those cell types. stClinic [34] employs a dynamic graph model with Gaussian mixture priors to delineate spatial niches across multi-slice multi-omics data, and uses attention-based learning to connect niche-level representations with clinical outcomes. While stClinic is powerful for population-scale niche discovery and label transfer, it fundamentally requires datasets where spatial and clinical information are matched—rare outside specialized studies—and its niche importance scores do not provide pathway-level mechanistic insights into why a niche is clinically aggressive or protective. STAN [31] estimates spot-specific transcription factor (TF) activities by integrating TF-target gene priors with spatial and morphological features via a spatially regularized kernel regression. STAN offers biologically grounded interpretability through known TF regulons and outperforms methods that ignore spatial context, but is restricted to transcription factor activity inference and does not link tissue regions to external clinical phenotypes or quantify broader gene pathway activities beyond TF-target relationships.

Critically, to the best of our knowledge, no existing ST analysis tool first leverages external bulk clinical phenotype data to identify phenotype-associated tissue domains in ST data, and then mechanistically characterizes those domains by quantifying the activity of known, biologically interpretable gene pathways within them. The ability to move from “which spatial regions are phenotype-associated” to “which molecular programs drive those associations” is essential for translating spatial findings into actionable biological insights and candidate therapeutic targets.

Existing unsupervised tissue domain segmentation tools can be broadly classified into non-deep-learning-based and deep-learning-based approaches. As representatives of the first category, BayesTME [30] and BayesSpace [33] detect spatial domains with similar gene expression patterns based on probabilistic graphical models. stLearn [22], as a deep-learning-based example, employs a convolutional neural network (CNN) to extract features from histological images, measuring morphological similarity between neighboring cells or spots and subsequently using these features to smooth gene expression before clustering.

SpaGCN [9], DeepST [27], STAGATE [4], and SEDR [28] use graph neural network (GNN) models to capture spatial, morphological, and expression-based relationships between spots, generating embedding representations for downstream clustering into spatial domains. However, these deep-learning-based methods lack biological interpretability due to their “black box” nature: the direct correspondence between input gene expression features, learned latent representations, and known biological processes remains opaque [19, 17]. Furthermore, none of these tools connect spatial domains to external clinical phenotypes, leaving the clinical relevance of identified domains uncharacterized.

To address these compounded limitations, we introduce GRAPHIST (**GR**Aph learning method for **P**henotype-based **I**nterpretable pathway activity identification of **S**patial **T**ranscriptomics), a graph learning framework that uniquely integrates three capabilities in a single pipeline. First, GRAPHIST integrates phenotype-associated bulk expression data with ST data by measuring similarity between each spatial spot and each bulk sample, constructing a correlation matrix that captures continuous phenotype relevance across the entire tissue without requiring matched spatial-clinical cohorts. Second, by optimizing a regression model on this correlation matrix, GRAPHIST detects tissue regions significantly associated with the phenotype of interest—going beyond unsupervised clustering to provide clinically guided spatial domain identification. Third, and distinctively from all existing methods, GRAPHIST employs a biologically interpretable variational graph autoencoder model to quantify the activity of prior gene pathways—not just cell types or transcription factors—in the identified phenotype-associated regions, highlighting the specific molecular programs that govern each phenotype association at the spatial level. When applied to pancreatic and breast cancer TME datasets, GRAPHIST successfully identified biologically relevant tissue regions tied to candidate biomarker gene knockdowns and survival outcomes, and pinpointed previously annotated pathways governing the corresponding phenotype associations in specific tissue spots, demonstrating its potential for uncovering pathway-level molecular insights within complex tumor microenvironments.

2 Methods

GRAPHIST operates as a sequential two-stage pipeline. In the first stage, it integrates phenotype-annotated bulk RNA-seq data with ST data to identify tissue spots significantly associated with a clinical phenotype of interest, using a spatially regularized sparse regression model. In the second stage, the identified phenotype-associated spots are characterized by a spatially-aware variational graph autoencoder with a pathway-masked decoder, which quantifies the activity of known biological pathways within those regions to reveal the molecular programs driving the phenotype associations. Both stages are described in detail below.

2.1 Phenotypic Bulk RNA-seq and ST Data Integration

GRAPHIST identifies phenotype-associated tissue spots by adapting the Scissor framework [25], originally developed for single-cell RNA-seq data, to the spatial transcriptomics setting. The key modification replaces the cell-cell similarity graph used in Scissor with a spot-spot spatial adjacency network, motivated by the biological intuition that physically proximate spots within a tissue share similar microenvironmental contexts and are therefore likely to exhibit similar associations with clinical phenotypes [20].

Bulk RNA-seq and ST Data Preprocessing Let $\mathbf{B} \in \mathbb{R}^{G \times M}$ denote the bulk RNA-seq gene expression matrix of M samples across G genes, with an associated phenotype vector $\mathbf{y} \in \mathbb{R}^M$ encoding the clinical outcome of interest (e.g., survival time, drug sensitivity, or binary treatment response). Let $\mathbf{E} \in \mathbb{R}^{G' \times N}$ denote the normalized gene expression matrix of N spatial spots across G' genes from the ST dataset. Prior to integration, the two expression matrices are restricted to their set of common genes, yielding aligned matrices $\mathbf{B}_c \in \mathbb{R}^{G_c \times M}$ and $\mathbf{E}_c \in \mathbb{R}^{G_c \times N}$ where G_c is the number of shared genes.

Spot-Bulk Similarity Matrix Construction To quantify the transcriptomic similarity between each spatial spot and each bulk sample, GRAPHIST computes a Pearson correlation matrix $\mathbf{C} \in \mathbb{R}^{N \times M}$, where each element C_{ij} is the Pearson correlation coefficient between the gene expression profile of spot i and bulk sample j across the G_c common genes:

$$C_{ij} = \frac{\sum_{g=1}^{G_c} (e_{gi} - \bar{e}_i) (b_{gj} - \bar{b}_j)}{\sqrt{\sum_{g=1}^{G_c} (e_{gi} - \bar{e}_i)^2} \sqrt{\sum_{g=1}^{G_c} (b_{gj} - \bar{b}_j)^2}}, \quad (1)$$

where e_{gi} and b_{gj} are the expression values of gene g in spot i and bulk sample j respectively, and $\bar{e}_i, \bar{b}_j$ are their respective means across genes. This correlation matrix $\mathbf{C}$ serves as the bridge between the phenotype-annotated bulk cohort and the spatial transcriptomics data, encoding how closely each spot’s transcriptional state resembles each bulk sample.

Spatial Spot-Spot Network Construction In the original Scissor method, a cell-cell similarity network is constructed from the single-cell data to regularize the regression problem and encourage cells with similar transcriptional profiles to receive similar phenotype association scores. In GRAPHIST, we replace this with a spatial adjacency network built directly from the tissue coordinates of the ST spots. A spatial neighborhood graph is constructed from the tissue coordinates of the ST spots and is shared across both stages of the GRAPHIST pipeline, though each stage uses a different form of the resulting adjacency matrix suited to its respective computational task. For each spot i , the n nearest spatial neighbors are identified by sorting pairwise Euclidean distances computed over the two-dimensional tissue coordinate space, and a binary adjacency

matrix $\mathbf{A} \in \{0, 1\}^{N \times N}$ is constructed such that $A_{ij} = 1$ if spot j is among the n nearest neighbors of spot i , and 0 otherwise. The matrix is then symmetrized as $\mathbf{A} \leftarrow (\mathbf{A} + \mathbf{A}^\top) > 0$ and self-loops are excluded. In Stage 1, this raw binary symmetric adjacency matrix $\mathbf{A}$ is used directly as the spatial network regularizer Ω in the regression optimization, encoding the constraint that spatially adjacent spots should receive similar phenotype association scores. In Stage 2, $\mathbf{A}$ is further processed into a normalized form suitable for graph convolution by applying symmetric degree normalization:

$$\hat{A} = D^{-1/2}(\mathbf{A} + I)D^{-1/2}, \quad (2)$$

where $\mathbf{D}$ is the degree matrix of $\mathbf{A} + I$ and I is the identity matrix, yielding the normalized adjacency $\hat{A}$ used throughout the GCN encoder and decoder computations. In all experiments, $n = 12$ nearest neighbors are used to define spatial adjacency.

This spatial network encodes the key assumption that physically neighboring spots within a tissue share similar microenvironmental contexts and therefore should exhibit similar phenotype association responses. This is a biologically motivated departure from Scissor’s transcriptome-based cell similarity network: while transcriptomic similarity captures molecular resemblance between cells regardless of their location, spatial adjacency captures the structural and microenvironmental continuity intrinsic to tissue organization, which we hypothesize is more appropriate for identifying spatially coherent phenotype-associated tissue domains.

Phenotype-Guided Regression with Spatial Regularization To identify spots whose transcriptional profiles are most correlated with the clinical phenotype, GRAPHIST optimizes a regression model on the spot-bulk correlation matrix $\mathbf{C}$ with respect to the phenotype vector $\mathbf{y}$, following the Scissor formulation [25]. A score vector $\boldsymbol{\alpha} \in \mathbb{R}^N$ is learned for all spots, where each α_i represents the degree of association between spot i and the phenotype of interest. The optimization problem incorporates three terms: a data fidelity term linking spot scores to the phenotype via the correlation matrix, an ℓ_1 sparsity penalty to identify a concise set of phenotype-relevant spots, and a spatial graph regularization term that encourages neighboring spots to receive similar scores:

$$\min_{\boldsymbol{\alpha}} \mathcal{L}(\mathbf{y}, \mathbf{C}^\top \boldsymbol{\alpha}) + \lambda_1 \|\boldsymbol{\alpha}\|_1 + \lambda_2 \boldsymbol{\alpha}^\top \mathbf{L} \boldsymbol{\alpha}, \quad (3)$$

where $\mathcal{L}$ is the loss function appropriate for the phenotype type (linear loss for continuous outcomes such as survival or drug sensitivity, logistic loss for binary outcomes such as treatment response or mutation status), $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is the graph Laplacian of the spatial adjacency network with degree matrix $\mathbf{D}$, and $\lambda_1, \lambda_2 \geq 0$ are regularization hyperparameters controlling sparsity and spatial smoothness respectively. The quadratic term $\boldsymbol{\alpha}^\top \mathbf{L} \boldsymbol{\alpha} = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} (\alpha_i - \alpha_j)^2$ penalizes large differences in phenotype association scores between spatially adjacent spots, promoting contiguous phenotype-associated tissue domains rather

than isolated individual spots. The optimization is solved using coordinate descent, and the hyperparameters λ_1 and λ_2 are selected via cross-validation on the bulk phenotype prediction task.

Phenotype-Associated Spot Identification After optimization, the regression coefficient vector $\alpha \in \mathbb{R}^N$ is extracted from the fitted model. Spots are categorized based on the sign of their learned coefficients: spots with $\alpha_i > 0$ are designated GRAPHIST+, corresponding to tissue regions whose transcriptional profiles resemble bulk samples associated with higher phenotype values (e.g., the knockdown condition, shorter survival, or treatment non-response), while spots with $\alpha_i < 0$ are designated GRAPHIST−, corresponding to phenotype-negative regions. Spots driven to exactly $\alpha_i = 0$ by the ℓ_1 penalty are classified as background.

To control the fraction of selected spots and avoid over-selection, the regularization hyperparameter α is chosen via an iterative search from a predefined candidate sequence of increasing values. For each candidate α , the optimal λ is determined by 10-fold cross-validation using 100 candidate λ values, retaining the value that minimizes the cross-validated loss (`lambda.min`). The selection rate, defined as the proportion of spots assigned to either GRAPHIST+ or GRAPHIST−:

$$\text{selection rate} = \frac{|\{\alpha_i > 0\}| + |\{\alpha_i < 0\}|}{N}, \quad (4)$$

is computed at each candidate α , and the smallest α for which the selection rate falls below a predefined cutoff of $\tau = 0.2$ is retained as the optimal parameter, ensuring that at most 20% of spots are selected with high confidence.

To further assess whether the identified phenotype-to-spot associations are statistically reliable rather than spurious, a permutation-based reliability significance test is applied following Scissor original paper instructions [25]. The test proceeds in three steps. First, k -fold cross-validation is performed on the spot-bulk correlation matrix $\mathbf{C}$, using training folds only to estimate α , and the prediction performance on held-out folds is evaluated to obtain an observed test statistic. The evaluation metric is chosen according to the phenotype type: mean squared error (MSE) for continuous outcomes, area under the ROC curve (AUC) for binary classification, and concordance index (C-index) for survival outcomes. Second, the bulk phenotype labels $\mathbf{y}$ are randomly permuted multiple times to sever the true phenotype-genotype relationship, and the same cross-validation procedure is repeated on each permuted dataset to build a null distribution of the evaluation metric. Finally, the observed test statistic is compared against this null distribution, and a permutation p -value is computed as the proportion of permutation-based statistics that equal or exceed (for AUC and C-index) or fall below (for MSE) the observed statistic. Only spot selections yielding $p < 0.05$ are considered statistically reliable and retained as GRAPHIST+ and GRAPHIST− regions. This test guards against false-positive associations that may arise when the bulk and ST datasets are poorly matched or when the correlation matrix carries insufficient phenotypic signal. The identified and statistically validated

GRAPHIST+ and GRAPHIST− spot sets are subsequently passed to Stage 2 of the GRAPHIST pipeline for pathway activity characterization.

2.2 Biologically Interpretable Pathway Activity Inference

To characterize the molecular programs operating within phenotype-associated tissue regions identified in the first stage, GRAPHIST employs a spatially-aware variational graph autoencoder with a biologically constrained decoder to infer the activity of known gene pathways at the spot level across the entire ST dataset. The model architecture, training objective, and downstream differential activity procedure are described below.

Spatial Graph Construction The normalized spatial adjacency matrix $\hat{A}$ used throughout the GCN encoder and decoder is derived from the same KNN spatial neighborhood graph constructed in Stage 1, as described in Section 2.1, via symmetric degree normalization of the binary adjacency matrix $\mathbf{A}$.

Pathway Mask Construction Biologically interpretable pathway representations are derived from a prior gene-pathway membership database provided in GMT file format (e.g. Reactome [10] or KEGG [12] database pathways). A binary mask matrix $M \in \{0, 1\}^{G \times P}$ is constructed, where G is the number of highly variable genes retained after preprocessing and P is the number of pathways, such that $M_{ij} = 1$ if gene i belongs to pathway j and 0 otherwise. One additional fully connected latent node is appended to M to capture residual transcriptional variance not attributable to any annotated pathway, yielding a final mask of dimension $G \times (P + 1)$. Only pathways with at least one member gene present in the filtered gene set are retained. The mask is held fixed throughout training and is not updated by backpropagation.

Model Architecture The biologically interpretable variational graph autoencoder model in GRAPHIST for inferring spot-based pathway activities consists of a graph convolutional encoder, a pathway-masked linear decoder, and a graph convolutional decoder for spatially smooth reconstruction, along with an inner product decoder for graph structure reconstruction. Below, we describe the technical implementation of each part in detail:

Encoder. The encoder is a two-layer graph convolutional network (GCN) [14]. Let $X \in \mathbb{R}^{N \times G}$ denote the normalized and scaled gene expression matrix of N spots across G genes, and let $\hat{A}$ denote the normalized spatial adjacency matrix. The first GCN layer maps the input to a hidden representation:

$$H = \text{ReLU}(\hat{A}XW_1), \quad (5)$$

where $W_1 \in \mathbb{R}^{G \times d_h}$ is a learnable weight matrix and $d_h = 800$ is the hidden dimension. Two separate second-layer GCN projections then produce the parameters of the variational posterior:

$$\mu = \hat{A}HW_\mu, \quad \log \sigma^2 = \hat{A}HW_\sigma, \quad (6)$$

where $W_\mu, W_\sigma \in \mathbb{R}^{d_h \times P'}$ and $P' = P + 1$ is the total number of latent pathway dimensions. The latent representation is obtained via the reparameterization trick:

$$z = \mu + \epsilon \odot \exp\left(\frac{1}{2} \log \sigma^2\right), \quad \epsilon \sim \mathcal{N}(0, I). \quad (7)$$

Each dimension of z thus corresponds to the inferred activity score of one known biological pathway (or the unannotated residual node) for each spot, informed by the expression of its spatial neighbors through the graph convolution operations.

Pathway-Masked Decoder. The primary decoder reconstructs gene expression from the pathway activity scores z using a sparse masked linear layer constrained by the pathway membership matrix M :

$$\hat{X} = z \cdot (W_{\text{dec}} \odot M^\top) + b, \quad (8)$$

where $W_{\text{dec}} \in \mathbb{R}^{P' \times G}$ is the decoder weight matrix, $\odot$ denotes element-wise multiplication, and b is a bias vector. The masking constraint enforces that each gene can only be reconstructed from the pathway latent nodes to which it biologically belongs, ensuring that learned pathway activities are grounded in their known gene membership. Furthermore, decoder weights are constrained to non-negative values throughout training by clamping $W_{\text{dec}} \geq 0$, ensuring that pathway activity scores are interpretable as activation levels rather than arbitrary signed quantities.

Graph Convolutional Decoder. A second, auxiliary GCN decoder reconstructs gene expression directly from z using the spatial adjacency:

$$\tilde{X} = \hat{A}zW_{\text{gc}}, \quad (9)$$

where $W_{\text{gc}} \in \mathbb{R}^{P' \times G}$ has a linear activation. This decoder encourages spatial coherence in the latent representations by penalizing reconstruction error with respect to the spatially smoothed expression profile.

Graph Structure Decoder. Following the standard variational graph autoencoder formulation [15], the spatial adjacency graph is reconstructed via inner products between latent spot representations. For each pair of spots (i, j) indexed in the adjacency mask, the reconstruction probability is:

$$\hat{A}_{ij} = \sigma(z_i \cdot z_j), \quad (10)$$

where σ is the sigmoid function. This term encourages spatially neighboring spots to have similar pathway activity profiles in the latent space.

Training Objective The model is trained end-to-end by minimizing a composite loss function combining three terms:

$$\mathcal{L} = \lambda_{\text{rec}} \cdot \mathcal{L}_{\text{rec}} + \lambda_{\text{gcn}} \cdot \mathcal{L}_{\text{gcn}} + \frac{\lambda_{\text{rec}}}{100} \cdot \mathcal{L}_{\text{rec,gcn}}, \quad (11)$$

where $\lambda_{\text{rec}} = 1000$ and $\lambda_{\text{gcn}} = 0.1$ in all experiments. The three component losses are defined as follows.

The *masked reconstruction loss* $\mathcal{L}_{\text{rec}}$ is the mean squared error (MSE) between the pathway-masked decoder output $\hat{X}$ and the input gene expression matrix X :

$$\mathcal{L}_{\text{rec}} = \frac{1}{NG} \|X - \hat{X}\|_F^2. \quad (12)$$

This loss trains the biologically constrained decoder to faithfully recover gene expression from pathway-structured latent codes.

The *graph reconstruction loss* $\mathcal{L}_{\text{gcn}}$ combines a binary cross-entropy term on the adjacency reconstruction with the KL divergence regularization of the variational posterior:

$$\mathcal{L}_{\text{gcn}} = \kappa \cdot \mathcal{L}_{\text{BCE}}(\hat{A}, A_{\text{mask}}) + D_{\text{KL}} \left(q(z | X, \hat{A}) \| p(z) \right), \quad (13)$$

where $p(z) = \mathcal{N}(0, I)$ is the standard Gaussian prior, $q(z | X, \hat{A}) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$ is the variational posterior parameterized by the GCN encoder, κ is a normalization constant proportional to the sparsity of the adjacency matrix [15], and A_{mask} is an augmented adjacency mask containing both true edges and an equal number of randomly sampled non-edges to form a balanced training signal for the graph reconstruction task. The KL divergence term expands analytically as:

$$D_{\text{KL}}(q \| p) = -\frac{1}{2N} \sum_{i=1}^N \sum_{k=1}^{P'} (1 + \log \sigma_{ik}^2 - \mu_{ik}^2 - \sigma_{ik}^2), \quad (14)$$

where μ_{ik} and σ_{ik}^2 are the mean and variance of the variational posterior for spot i and pathway dimension k , as produced by the GCN encoder [13].

The *GCN reconstruction loss* $\mathcal{L}_{\text{rec,gcn}}$ is the MSE between the auxiliary GCN decoder output $\tilde{X}$ and the input X :

$$\mathcal{L}_{\text{rec,gcn}} = \frac{1}{NG} \|X - \tilde{X}\|_F^2. \quad (15)$$

This term is downweighted by a factor of 100 relative to $\mathcal{L}_{\text{rec}}$, serving as a spatial regularizer that encourages local smoothness in the pathway latent space without dominating the biologically constrained reconstruction objective.

All models are optimized using the Adam optimizer with learning rate 10^{-4} and weight decay 5×10^{-4} . Training runs for 100 epochs on the breast cancer datasets and 200 epochs on the pancreatic cancer dataset.

Preprocessing For each ST dataset, raw count matrices are loaded and genes expressed in fewer than 50 spots or with fewer than 10 total counts are removed. Library sizes are normalized to 10^6 counts per spot, and the top 5,000 highly variable genes are selected using the Seurat v3 criterion [24]. Gene expression values are then z-score scaled to zero mean and unit variance across spots. For dimensionality reduction used in downstream clustering, PCA is applied to the scaled matrix using 200 components with a fixed random seed via `sklearn`.

Differential Pathway Activity Between Phenotype-Associated Regions

To identify pathways that are differentially active between phenotype-associated spots identified in Stage 1 and background spots, GRAPHIST employs a Bayesian differential activity procedure adapted from scVI [16]. Let $\mathcal{S}_1$ denote the set of phenotype-positive spots (GRAPHIST+) and $\mathcal{S}_2$ denote the phenotype-negative spots (GRAPHIST-) as returned by the regression model in Stage 1.

For each group, $n_{\text{samples}} = 5,000$ spots are drawn with replacement from $\mathcal{S}_1$ and $\mathcal{S}_2$ respectively, and their pathway activity vectors are retrieved from Z . To better estimate the double integral underlying the Bayesian hypothesis test, $n_{\text{perm}} = 5,000$ random cross-group pairs are sampled via a permutation procedure [16], yielding matched activity arrays $z_1, z_2 \in \mathbb{R}^{n_{\text{perm}} \times P'}$.

For each pathway k , the Bayes factor is computed as:

$$\text{BF}_k = \log \frac{p(\hat{z}_{1k} > \hat{z}_{2k})}{p(\hat{z}_{1k} \leq \hat{z}_{2k})} = \log \frac{p_{H_1} + \epsilon}{p_{H_2} + \epsilon}, \quad (16)$$

where $p_{H_1} = \frac{1}{n_{\text{perm}}} \sum \mathbf{1}[z_{1k} > z_{2k}]$ is the posterior probability that pathway k is more active in $\mathcal{S}_1$ than in $\mathcal{S}_2$, $p_{H_2} = 1 - p_{H_1}$, and $\epsilon = 10^{-12}$ is a numerical stability constant. The mean difference:

$$\Delta_k = \frac{1}{n_{\text{perm}}} \sum_{j=1}^{n_{\text{perm}}} (z_{1k}^{(j)} - z_{2k}^{(j)}) \quad (17)$$

serves as the effect size metric, capturing the magnitude and direction of differential activity for pathway k between the two spot groups. Pathways are ranked by BF_k and visualized as volcano plots displaying $\log_e(\text{BF}_k)$ against $|\Delta_k|$, with significance thresholds applied to both axes to identify the most differentially active pathways.

2.3 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Run-in Heading in Bold. Text follows	10 point, bold
4th-level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{18}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

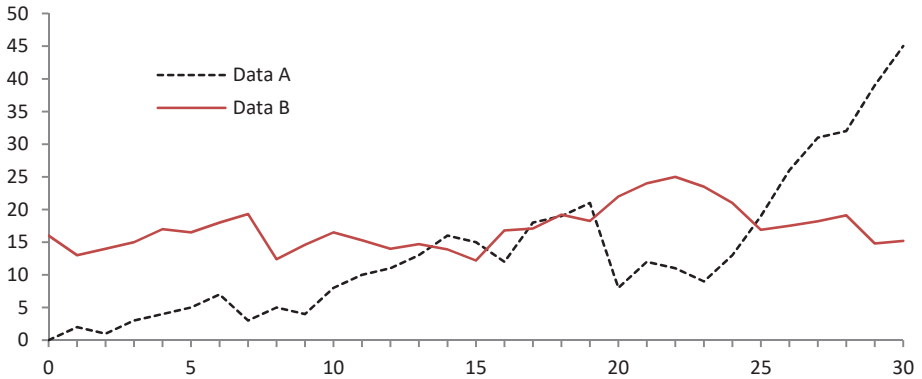


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries

for journal articles [?], an LNCS chapter [?], a book [?], proceedings without editors [?], and a homepage [?]. Multiple citations are grouped [?,?,?], [?,?,?,?].

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References

1. Arora, R., Cao, C., Kumar, M., Sinha, S., Chanda, A., McNeil, R., Samuel, D., Arora, R.K., et al: Spatial transcriptomics reveals distinct and conserved tumor core and edge architectures that predict survival and targeted therapy response. *Nat Commun* **14**(5029) (2023). <https://doi.org/10.1038/s41467-023-40271-4>
2. Cheng, X., Tang, C., Dong, K., et al: Spalinker identifies phenotype-associated spatial tumor microenvironment features by integrating bulk and spatial sequencing data. *Cell Genom* (2025). <https://doi.org/10.1016/j.xgen.2025.100893>
3. Clough, E., Barrett, T.: The gene expression omnibus database. *Methods Mol Biol* **1418**, 93–110 (2016). https://doi.org/10.1007/978-1-4939-3578-9_5
4. Dong, K., Zhang, S.: Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nat Commun* **13**(1739) (2022). <https://doi.org/10.1038/s41467-022-29439-6>
5. Duan, B., Cheng, X., Zou, H.: Spapheno: linking spatial transcriptomics to clinical phenotypes with interpretable machine learning. *Genome Med* **18**(67) (2026). <https://doi.org/10.1186/s13073-026-01645-7>
6. Echarti, A., Hecht, M., Büttner-Herold, M., Haderlein, M., Hartmann, A., Fietkau, R., Distel, L.: Cd8+ and regulatory t cells differentiate tumor immune phenotypes and predict survival in locally advanced head and neck cancer. *Cancers* **11**(9) (2019). <https://doi.org/10.3390/cancers11091398>
7. Gajewski, T.F., Woo, S.R., Zha, Y., Spaapen, R., Zheng, Y., Corrales, L., Spranger, S.: Cancer immunotherapy strategies based on overcoming barriers within the tumor microenvironment. *Current Opinion in Immunology* **25**(2), 268–276 (2013). <https://doi.org/10.1016/j.coi.2013.02.009>
8. Hu, J., Coleman, K., Zhang, D., Lee, E.B., Kadara, H., Wang, L., Li, M.: Deciphering tumor ecosystems at super resolution from spatial transcriptomics with tesla. *Cell Systems* **14**(5) (2023). <https://doi.org/10.1016/j.cels.2023.03.008>
9. Hu, J., Li, X., Coleman, K., Schroeder, A., Ma, N., Irwin, D.J., Lee, E.B., Shinohara, R.T., Li, M.: Spagcn: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional

³ If EquinOCS, our proceedings submission system, is used, then the disclaimer can be provided directly in the system.

- network. *Nat Methods* **18**, 1342–1351 (2021). <https://doi.org/10.1038/s41592-021-01255-8>
10. Jassal, B., Matthews, L., Viteri, G., Gong, C., Lorente, P., Fabregat, A., Sidiropoulos, K., Cook, J., Gillespie, M., Haw, R., Loney, F., May, B., Milacic, M., Rothfels, K., Sevilla, C., Shamovsky, V., Shorser, S., Varusai, T., Weiser, J., Wu, G., Stein, L., Hermjakob, H., D'Eustachio, P.: The reactome pathway knowledgebase. *Nucleic acids research* **48**(D1), D498–D503 (2020). <https://doi.org/10.1093/nar/gkz1031>
 11. Jin, Y., Zuo, Y., Li, G., Liu, W., Pan, Y., Fan, T., Fu, X., Yao, X., Peng, Y.: Advances in spatial transcriptomics and its applications in cancer research. *Mol Cancer* **23**(129) (2024). <https://doi.org/10.1186/s12943-024-02040-9>
 12. Kanehisa, M., Goto, S.: Kegg: kyoto encyclopedia of genes and genomes. *Nucleic acids research* **28**(1), 27–30 (2000)
 13. Kingma, D.P., Welling, M.: Auto-encoding variational bayes. In: 2nd International Conference on Learning Representations, ICLR 2014 (2014), <https://arxiv.org/abs/1312.6114>
 14. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. *arXiv:1609.02907v4* (2016)
 15. Kipf, T.N., Welling, M.: Variational graph auto-encoders. *arXiv:1611.07308v1* (2016)
 16. Lopez, R., Regier, J., Cole, M.B., Jordan, M.I., Yosef, N.: Deep generative modeling for single-cell transcriptomics. *Nature methods* **15**(12), 1053–1058 (2018)
 17. Lopez, R., Tagasovska, N., Ra, S., Cho, K., Pritchard, J., Regev, A.: Learning causal representations of single cells via sparse mechanism shift modeling **213**, 662–691 (11–14 Apr 2023), <https://proceedings.mlr.press/v213/lopez23a.html>
 18. Maffuid, K., Cao, Y.: Decoding the complexity of immune-cancer cell interactions: Empowering the future of cancer immunotherapy. *Cancers* **15**(16) (2023). <https://doi.org/10.3390/cancers15164188>
 19. Niyakan, S., Luo, X., Yoon, B.J., Qian, X.: Biologically interpretable vae with supervision for transcriptomics data under ordinal perturbations. *ICLR 2024 Workshop on Machine Learning for Genomics Explorations* (2024)
 20. Niyakan, S., Qian, X.: Phenograph: A multi-agent framework for phenotype-driven discovery in spatial transcriptomics data augmented with knowledge graphs. *bioRxiv* 2025.06.06.658341 (2025)
 21. Niyakan, S., Sheng, J., Cao, Y., Zhang, X., Xu, Z., Wu, L., Wong, S.T., Qian, X.: Mustang: Multi-sample spatial transcriptomics data analysis with cross-sample transcriptional similarity guidance. *Patterns* **5**(5) (2024). <https://doi.org/10.1016/j.patter.2024.100986>
 22. Pham, D., Tan, X., Balderson, B., Xu, J., Grice, L.F., Yoon, S., Willis, E.F., Tran, M., et al: Robust mapping of spatiotemporal trajectories and cell–cell interactions in healthy and diseased tissues. *Nat Commun* **14**(7739) (2023). <https://doi.org/10.1038/s41467-023-43120-6>
 23. Ruan, X., Cheng, Y., Ye, Y., Wang, Y., Chen, X., Yang, Y., Liu, T., Yan, F.: Pipet: predicting relevant subpopulations in single-cell data using phenotypic information from bulk data. *Briefings in Bioinformatics* **25**(4) (2024). <https://doi.org/10.1093/bib/bbae260>
 24. Stuart, T., Butler, A., Hoffman, P., et al.: Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019). <https://doi.org/10.1016/j.cell.2019.05.031>
 25. Sun, D., Guan, X., Moran, A.E., Wu, L.Y., Qian, D.Z., et al.: Identifying phenotype-associated subpopulations by integrating bulk and single-cell sequencing data. *Nat Biotechnol* **40**, 93–110 (2022). <https://doi.org/10.1038/s41587-021-01091-3>

26. Weinstein, J.N., Collisson, E.A., Mills, G.B., et al.: The cancer genome atlas pan-cancer analysis project. *Nat Genet* **45**, 1113–1120 (2013). <https://doi.org/10.1038/ng.2764>
27. Xu, C., Jin, X., Wei, S., Wang, P., Luo, M., Xu, Z., Yang, W., Cai, Y., Xiao, L., et al: Deepst: identifying spatial domains in spatial transcriptomics by deep learning. *Nucleic Acids Research* **50**(22), e131 (2022). <https://doi.org/10.1093/nar/gkac901>
28. Xu, H., Fu, H., Long, Y., Ang, K.S., Sethi, R., Chong, K., Li, M., Uddamvathanak, R., Lee, H.K., Ling, J., Chen, A., Shao, L., Liu, L., Chen, J.: Unsupervised spatially embedded deep representation of spatial transcriptomics. *Genome Medicine* **16**(12) (2024). <https://doi.org/10.1186/s13073-024-01283-x>
29. Yang, W., Soares, J., Greninger, P., Edelman, E.J., Lightfoot, H., Forbes, S., et al.: Genomics of drug sensitivity in cancer (gdsc): a resource for therapeutic biomarker discovery in cancer cells. *Nucleic acids research* **41** (2013). <https://doi.org/10.1093/nar/gks1111>
30. Zhang, H., Hunter, M.V., Chou, J., Quinn, J.F., et al.: Bayestme: An end-to-end method for multiscale spatial transcriptional profiling of the tissue microenvironment. *Cell systems* **14**(7), 605–619 (2023), <https://doi.org/10.1016/j.cels.2023.06.003>
31. Zhang, L., Sagan, A., Qin, B., et al: Stan, a computational framework for inferring spatially informed transcription factor activity. *Nucleic Acids Research* **54** (2026). <https://doi.org/10.1093/nar/gkaf1473>
32. Zhang, Q., Jin, S., Zou, X.: scab detects multiresolution cell states with clinical significance by integrating single-cell genomics and bulk sequencing data. *Nucleic acids research* **50**(21), 12112–12130 (2022). <https://doi.org/10.1093/nar/gkac1109>
33. Zhao, E., Stone, M.R., Ren, X., Guenthoer, J., Smythe, K.S., Pulliam, T., Williams, S.R., Uytingco, C.R., et al.: Spatial transcriptomics at subspot resolution with bayesspace. *Nat Biotechnol* **39**, 1375–1384 (2021). <https://doi.org/10.1038/s41587-021-00935-2>
34. Zuo, C., Xia, J., Xu, Y., et al: stclinic dissects clinically relevant niches by integrating spatial multi-slice multi-omics data in dynamic graphs. *Nat Commun* **16** (2025). <https://doi.org/10.1038/s41467-025-60575-x>